▸ JHON PEREZ, CARLOS UZCÁTEGUI, *Complexity of topological conjugacy in countable dynamical systems.*

Escuela de Matemáticas, Universidad Industrial de Santander, Bucaramanga, Colombia.
E-mail: `prjhonf@gmail.com`.

A dynamical system is a pair (X, f), where X is a compact metric space and $f : X \to X$ is a continuous function. If moreover, X is countable, we say that (X, f) is a countable dynamical system. Two dynamical systems (X, f) and (Y, g) are said to be topologically conjugate if there exists a homeomorphism $\varphi : X \to Y$ such that $\varphi \circ f = g \circ \varphi$. When the phase space X is fixed, topological conjugacy induces an equivalence relation on the space of continuous self-maps of X.

The aim of this talk is to present a tool from Descriptive Set Theory, known as Borel reducibility, which is useful for classifying equivalence relations on Polish spaces. In particular, we focus on the study of the descriptive complexity of topological conjugacy in dynamical systems. We review several results from the literature concerning the cases where the phase space is the interval or the Cantor set [1, 2, 3]. Finally, we present some recent results of our own related to the complexity of conjugacy in countable dynamical systems. More precisely, our results extend, in part, a result of Kaya [4] on dynamical systems with a dense orbit.

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[4] Kaya B, *The complexity of topological conjugacy of pointed Cantor minimal systems*, ***Archive for Mathematical Logic***, vol. 56 (2017), no. 1, pp. 215–235.